

1 Integral Recurrent Sequences

题 1.1. (China 2020) Let $a_1 = 1, a_2 = 2, a_n = 2a_{n-1} + a_{n-2}, n = 3, 4, \dots$. Prove that for any integer $n \geq 5$, a_n has at least one prime factor p , such that $p \equiv 1 \pmod{4}$.

题 1.2. (Korea MO 2013) Two coprime positive integers a, b are given. Integer sequence $\{a_n\}, \{b_n\}$ satisfies

$$(a + b\sqrt{2})^{2n} = a_n + b_n\sqrt{2}$$

Find all prime numbers p such that there exist positive integer $n \leq p$ satisfying $p \mid b_n$.

题 1.3. Consider the sequence $a_0 = 0, a_1 = 1$ and $a_n = 2a_{n-1} + a_{n-2}$. Prove that $\frac{a_n}{n}$, written as an irreducible fraction, has both numerator and denominator odd.

题 1.4. (China Round2 2022) $k > 2$ is an integer. $a_0, a_1, \dots$ is an integer sequence such that $a_0 = 0, a_{n+1} = ka_n - a_{n-1}$. Prove that for any positive integer m , $(2m)! \mid a_1 a_2 \dots a_{3m}$.

题 1.5. (ELMO 2020) Define the Fibonacci numbers by $F_1 = F_2 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 3$. Let k be a positive integer. Suppose that for every positive integer m there exists a positive integer n such that $m \mid F_n - k$. Must k be a Fibonacci number?

2 Graph Theory and Applications

题 2.1. Can one put $1, 2, \dots, 9$ on a circle such that sum of every two adjacent numbers is not divisible by 3, 5 or 7?

题 2.2. Split the set $\{1, 2, \dots, 12\}$ into 6 disjoint subsets of 2 elements, such that the two elements in each subset are coprime. Find the number of such splittings.

题 2.3. (Canada 1988) A host is waiting for arrival of either 7 or 11 boys. He prepares 77 marbles as gifts. He split the marbles into n bags such that each boy will receive some bags and all 77 marbles will be equally distributed among the boys, no matter 7 or 11 boys will arrive. Find the least number n such that this could happen.

题 2.4. Given coprime positive integers a, b . Find the largest integer n , such that there exists sequence of real numbers $a_1, a_2, \dots, a_n$, every consecutive a terms of the sequence has a positive sum, while every consecutive b terms of the sequence has a negative sum.

题 2.5. (IMOSL2002) Let n be an even positive integer. Show that there is a permutation $(x_1, x_2, \dots, x_n)$ of $(1, 2, \dots, n)$ such that for every $i \in \{1, 2, \dots, n\}$, the number x_{i+1} is one of the numbers $2x_i, 2x_i - 1, 2x_i - n, 2x_i - n - 1$. Hereby, we use the cyclic subscript convention, so that x_{n+1} means x_1 .

题 2.6. (IMO 2020) There are $4n$ pebbles of weights $1, 2, 3, \dots, 4n$. Each pebble is coloured in one of n colours and there are four pebbles of each colour. Show that we can arrange the pebbles into two piles so that the following two conditions are both satisfied:

- The total weights of both piles are the same.
- Each pile contains two pebbles of each colour.